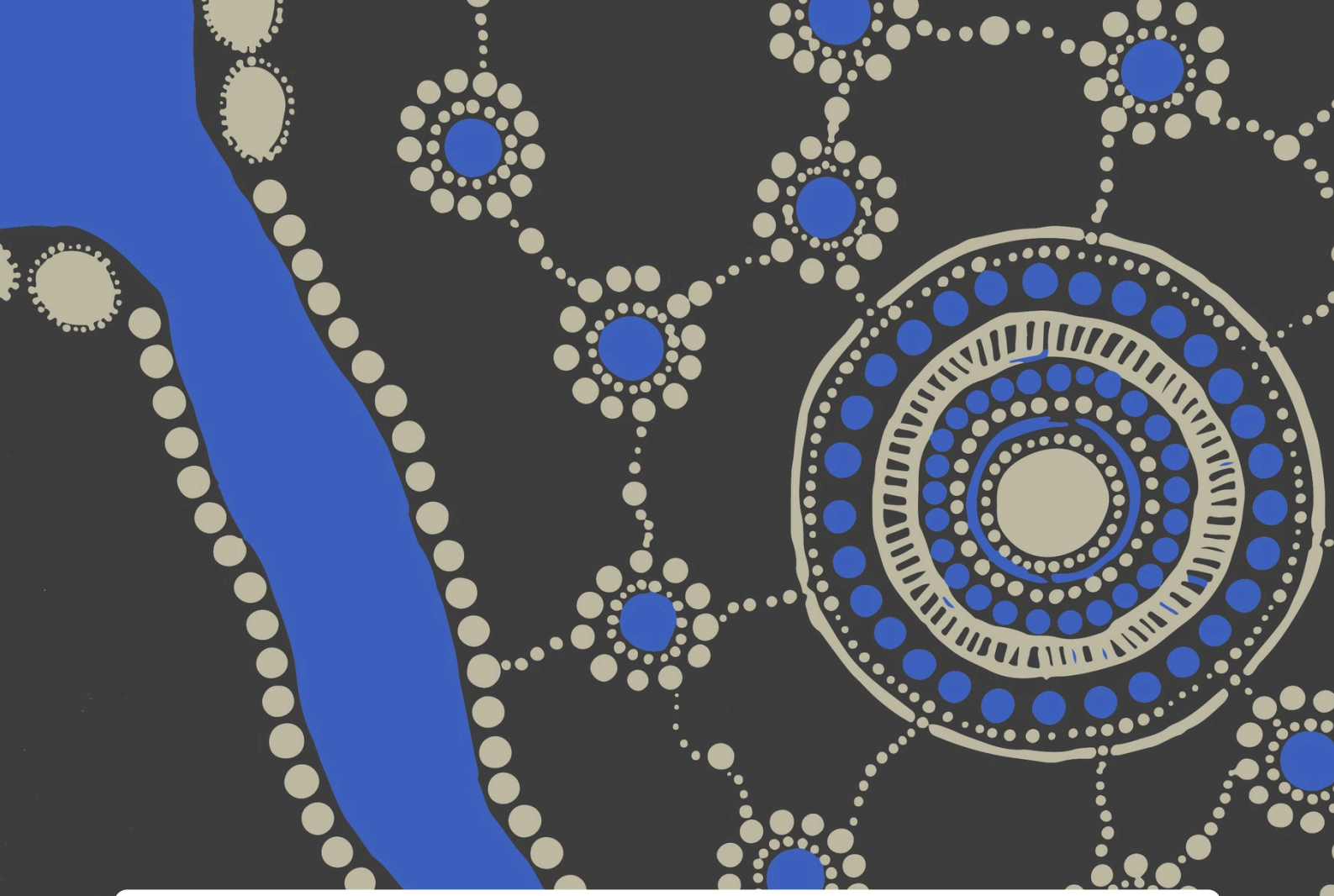


KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

Mathematics Standard 2
Year 12

CHAPTER 1 · KEY IDEAS AND REVIEW

Ratios and rates

Year 12 Standard 2 • Simplified review workbook

NSW Mathematics Standard 11–12 Syllabus (2024) • Outcomes MST-12-S2-05, MAO-WM-01

Name _____ Class _____ Teacher _____

Cover and banner artwork: *Kingscliff High School Vision Statement* by local Indigenous artist Marni Tuala.

About this workbook

This workbook is a simplified version of the **Chapter 1 key ideas and review** from your textbook, *CambridgeMATHS Mathematics Standard 2 Year 12*. There is one lesson for each key idea, and each lesson has three parts:

Key idea: the idea in plain words, with the *I can* statements from the textbook checklist.

Worked examples: every example is fully worked, with the reason for each line next to it. The answer is in a blue box.

Your turn: **multiple choice**, **short answer** and **extended response** questions, like the textbook review. Each question tells you which example to look at.

The workbook ends with a **chapter review** set out the same way, and answers to every question.

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WORKING ON YOUR OWN?

1. Read the key idea and look at the picture.
2. Read a worked example one line at a time. Cover the next line: can you say what comes next?
3. Try the questions that point to that example. Write every step in the box.
4. Check the answers at the back, fix any mistakes, then rate yourself on the checklist.

Syllabus outcomes

MST-12-S2-05	solves practical problems involving ratios and rates
MAO-WM-01	develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly

Formulas you'll use (some are on the HSC reference sheet)

Trapezoidal rule: $A \approx \frac{h}{2}(d_f + d_l)$ Speed: $S = \frac{D}{T}$ Volume: $V = Ah$, $1\ \text{m}^3 = 1000\ \text{L}$

Fuel used (L) = $\frac{\text{distance (km)}}{100} \times$ rate (L/100 km) Energy (kWh) = power (kW) × time (h)

Acknowledgement of Country

We acknowledge the Traditional Custodians of the land on which we learn, the Bundjalung people, and pay our respects to Elders past and present. The artwork on the cover and throughout this workbook is our school vision statement, created by local Indigenous artist Marni Tuala.

You will need

A calculator, a ruler and a pencil. Show your working in every box: in the HSC, working earns marks even when the final answer is wrong.

Key ideas Chapter 1 at a glance

A simplified version of the textbook's *Key ideas and chapter summary*. Each key idea has its own lesson.

			EXAMPLE
1	Ratio	Compares amounts of the **same kind**, in a set order. Multiply or divide every part by the same number to get an **equivalent ratio**.	5 : 2 is not the same as 2 : 5 5 : 2 = 10 : 4 = 15 : 6
2	Capture–recapture	Estimates the size of an animal population. Tag some animals, release them, then catch a second sample.	$\frac{\text{tagged in 2nd sample}}{\text{size of 2nd sample}} = \frac{\text{tagged at first}}{\text{population}}$
3	Dividing a quantity in a ratio	**1** Add the parts (2 + 3 = 5). **2** Write each part over the total. **3** Multiply the quantity by each fraction.	Share \$40 in the ratio 2 : 3. $\frac{2}{5} \times \$40 = \$16$, $\frac{3}{5} \times \$40 = \$24$
4	Scale drawings	Scale = drawing length : actual length. A scale can have units (1 cm to 1 m) or no units (1 : 100).	Scale 1 : 100. 3 cm on the plan → real length = 3 × 100 = 300 cm = 3 m
5	Plans and elevations	A **plan** is the view from the top. An **elevation** is the view from one side, such as the front or the side.	plan = bird's-eye view front elevation = view from the front
6	Perimeter and area of land	**Perimeter**: measure each side on the photo and multiply by the scale. **Area**: count grid squares to estimate, or use a formula. For one curved side, use the trapezoidal rule.	$A \approx \frac{h}{2}(d_f + d_l)$ h = distance between the two measurements, d_f = first, d_l = last
7	Volume of water	If the depth is the same everywhere, volume = area of the surface × depth. For rain, change millimetres to metres first.	$V = Ah$ (h = depth), 1 m^3 = 1000 L 18 mm = 0.018 m
8	Rates	Compares two amounts with **different units**.	km/h, \$/kg, L/100 km, \$/hour 25 mL/min = 25 × 60 mL/h = 1500 mL/h = 1.5 L/h
9	Unitary method	Divide to find the amount for **1**. Multiply to find the amount you need. For the best buy, compare prices for the same amount.	3 kg costs \$12 1 kg: \$12 ÷ 3 = \$4 → 5 kg: 5 × \$4 = \$20
10	Speed	Speed is a rate. On a distance–time graph, a steeper line is faster and a flat line means stopped.	$S = \frac{D}{T}$, $D = S \times T$
11	Energy	Energy (kWh) = power (kW) × time (hours). Running cost = energy × price per kWh.	1 kW = 1000 W, so 600 W = 0.6 kW 2 kW for 3 h = 6 kWh at \$0.30 per kWh: 6 × \$0.30 = \$1.80
12	Fuel consumption	A rate given in litres per 100 km (L/100 km) or km per litre.	Fuel used (L) = $\frac{\text{distance (km)}}{100} \times \text{rate}$ Rate = $\frac{\text{litres}}{\text{km}} \times 100$ 30 L for 400 km → 7.5 L/100 km

1 Ratios

Key idea: Ratio • Textbook 1A • MST-12-S2-05

Key idea Ratio

A **ratio** compares two or more amounts of the **same kind**, in a set order.

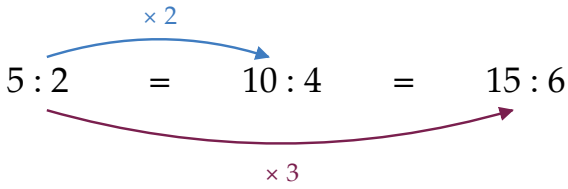
- A builder mixes 5 buckets of sand with 2 buckets of cement. The ratio of sand to cement is 5 : 2, read "5 to 2".
- **Order matters.** The ratio of cement to sand is 2 : 5.
- A ratio has **no units**. If the units are different, change them to the same unit first.



sand : cement = 5 : 2

Equivalent ratios

Multiply or divide **every** part by the **same number**. The ratio stays the same, just like equivalent fractions.



Simplest form

Divide every part by the **highest common factor** (HCF): the biggest number that divides into every part. A ratio is in simplest form when the parts are whole numbers and only 1 divides into all of them.

- **Decimals?** Multiply every part by 10 (or 100) first to make whole numbers.
- **Different units?** Change to the same unit first, e.g. 40 cm : 2 m = 40 cm : 200 cm = 40 : 200 = 1 : 5.

Three ways to write a ratio

Form	How to get it	Example: 8 : 10
Simplest form	Divide every part by the HCF.	8 : 10 = 4 : 5
The form 1 : a	Divide every part by the **first** number.	8 ÷ 8 : 10 ÷ 8 = 1 : 1.25
The form b : 1	Divide every part by the **second** number.	8 ÷ 10 : 10 ÷ 10 = 0.8 : 1

Finding an unknown

Find two numbers in the same place. Divide to find what one was multiplied by. Then do the **same** to the other part.

$$3 : 4 = a : 12 \quad \rightarrow \quad 12 \div 4 = \mathbf{3}, \text{ so } a = 3 \times \mathbf{3} = 9$$

Hint **Finding the HCF.** Take the smallest number. Try its factors from biggest to smallest until one also divides into every other number. For 12 : 18: does 12 go into 18? No. Does 6? Yes, so the HCF is 6.

CHECKLIST · SUCCESS CRITERIA

- ▸ *I can* change a ratio to simplest form, to the form 1 : a and to the form b : 1. ○○○
- ▸ *I can* use a ratio to find an unknown amount. ○○○

Rate yourself when you finish the lesson: ○○○ not yet / getting there / got it

WORKED EXAMPLE 1.1 — Simplest form

Write each ratio in simplest form. (a) $12:18$ (b) $10:15:30$ (c) $4.8:1.6$

	WORKING	WHY
(a)	$12:18$	The HCF of 12 and 18 is **6**.
	$= 12 \div 6 : 18 \div 6$	Divide **both** parts by 6.
	$= \boxed{2:3}$	Only 1 divides into 2 and 3, so this is simplest form.
(b)	$10:15:30$	The HCF of 10, 15 and 30 is **5**.
	$= 10 \div 5 : 15 \div 5 : 30 \div 5$	Divide **every** part by 5.
	$= \boxed{2:3:6}$	Only 1 divides into 2, 3 and 6.
(c)	$4.8:1.6$	There are decimals. Make whole numbers first.
	$= 4.8 \times 10 : 1.6 \times 10$	Multiply **both** parts by **10**.
	$= 48:16$	The HCF of 48 and 16 is **16**.
	$= 48 \div 16 : 16 \div 16$	Divide both parts by 16.
	$= \boxed{3:1}$	Keep the 1. Write 3:1, not just 3.

WORKED EXAMPLE 1.2 — Three ways to write a ratio

Consider the ratio $15:12$. Write it (a) in simplest form (b) in the form $1:a$ (c) in the form $b:1$.

	WORKING	WHY
(a)	$15:12$	The HCF of 15 and 12 is **3**.
	$= 15 \div 3 : 12 \div 3$	Divide both parts by 3.
	$= \boxed{5:4}$	
(b)	$15:12$	To make the **first** part 1, divide both parts by the first number, 15.
	$= 15 \div 15 : 12 \div 15$	Use your calculator: 12 ÷ 15 = 0.8
	$= \boxed{1:0.8}$	The first part is now 1.
(c)	$15:12$	To make the **second** part 1, divide both parts by the second number, 12.
	$= 15 \div 12 : 12 \div 12$	Use your calculator: 15 ÷ 12 = 1.25
	$= \boxed{1.25:1}$	The second part is now 1.

WORKED EXAMPLE 1.3 — Finding an unknown

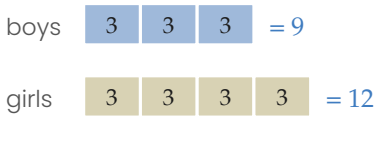
Find the value of the pronumeral. (a) $2:5 = a:20$ (b) $3:4 = 15:b$

	WORKING	WHY
(a)	$2:5 = a:20$	Find two numbers in the same place: 5 and 20 are both **second** parts.
	$20 \div 5 = \mathbf{4}$	Divide to find what 5 was multiplied by ($5 \times 4 = 20$).
	$a = 2 \times \mathbf{4}$	Do the same to the first part.
	$\boxed{a = 8}$	Check: 8:20 simplifies to 2:5.
(b)	$3:4 = 15:b$	3 and 15 are both **first** parts.
	$15 \div 3 = \mathbf{5}$	Divide to find what 3 was multiplied by ($3 \times 5 = 15$).
	$b = 4 \times \mathbf{5}$	Do the same to the second part.
	$\boxed{b = 20}$	Check: 15:20 simplifies to 3:4.

WORKED EXAMPLE 1.4 — Using a ratio to find an amount

The ratio of boys to girls in a class is 3 : 4. There are 12 girls. How many boys are in the class?

WORKING	WHY
boys : girls = 3 : 4	Write the ratio with labels so you know which part is which.
4 parts = 12 girls	Girls are the **second** part, so 4 parts = 12.
1 part = 12 ÷ 4 = 3	Find **one part** first.
boys = 3 parts = 3 × 3	Boys are the **first** part, 3 parts.
= 9 boys	Check: 9 : 12 simplifies to 3 : 4.



YOUR TURN 1

Stuck? Each question shows the worked example to look at.

Multiple choice circle the best answer

1 Which ratio is equivalent to 2 : 5?

→ Equivalent ratios

A. 4 : 10 **B.** 5 : 2
C. 4 : 7 **D.** 2 : 10

2 What is 15 : 20 in simplest form? → Example 1.1

A. 5 : 4 **B.** 3 : 4
C. 1 : 5 **D.** 30 : 40

3 The ratio of adults to children in a park is 5 : 9. There are 630 children in the park. How many adults are in the park? *Hint: 9 parts = 630 children.* → Example 1.4

A. 70 **B.** 126 **C.** 280 **D.** 350

Short answer show your working in the box

1 Write each ratio in simplest form. *Hint: find the HCF first.* → Example 1.1

(a) 20 : 30 (b) 6 : 9 : 12 (c) 1.5 : 2.5

(a)	(b)	(c)

2 Consider the ratio 10 : 4. Write it → Example 1.2

(a) in simplest form (b) in the form 1 : a (c) in the form b : 1

(a)	(b)	(c)

3 Find the value of the pronumeral. *Hint: divide to find what was multiplied.* → Example 1.3

(a) $2 : 3 = a : 15$ (b) $4 : 3 = a : 24$ (c) $3 : 5 = 12 : b$

(a)	(b)	(c)

4 Gabriel mixes sand and cement in the ratio 5 : 2. He uses 4 buckets of cement. How many buckets of sand should he use? *Hint: 2 parts = 4 buckets.* → Example 1.4

5 The ratio of Victoria's height to Willow's height is 8 : 7. Victoria is 176 cm tall. How tall is Willow? *Hint: 8 parts = 176 cm.* → Example 1.4

Extended response answer each part in order

1 A fruit punch recipe mixes orange juice and lemonade in the ratio 3 : 5. → Key idea and Example 1.4

(a) Write the ratio of **lemonade** to **orange juice**.

(b) Mia uses 600 mL of orange juice. How many millilitres is **one part**?

(c) How much lemonade does Mia need?

(d) How much punch does Mia make altogether? *Hint: add the orange juice and the lemonade.*

Finished? Check your answers in the Answers section at the back. Then go back to the checklist on page 4 and rate yourself.

Answers

Lesson 1 Ratios

MULTIPLE CHOICE

1 A **2** B

3 D (1 part = 630 ÷ 9 = 70, adults = 5 × 70 = 350)

SHORT ANSWER

1 (a) 2 : 3 (b) 2 : 3 : 4 (c) 3 : 5

2 (a) 5 : 2 (b) 1 : 0.4 (c) 2.5 : 1

3 (a) $a = 10$ (b) $a = 32$ (c) $b = 20$

4 1 part = 4 ÷ 2 = 2,
sand = 5 × 2 = 10 buckets

5 1 part = 176 ÷ 8 = 22 cm,
Willow = 7 × 22 = 154 cm

EXTENDED RESPONSE

1 (a) 5 : 3
(b) 600 ÷ 3 = 200 mL
(c) 5 × 200 = 1000 mL (1 L)
(d) 600 + 1000 = 1600 mL (1.6 L)